Keyword Research MCP

Design overview and intended use — a local Model Context Protocol server for normalized Google Ads keyword-planning data

Purpose

Keyword Research MCP is a single-user, locally run server that makes Google Ads keyword-planning data available to an AI assistant during content and search planning. It answers one question: what do people actually search for in a given topic and market, and how much? The operator runs it on their own machine against their own Google Ads account; there is no hosted service, no multi-tenant deployment, and no redistribution of Google Ads data to third parties.

Use case

A practitioner planning website content asks their assistant to research a topic. The assistant calls this server to retrieve keyword ideas and demand figures, then reasons over the results — grouping related phrases, judging intent, and proposing which pages are worth writing. Without it, that planning rests on guesswork; with it, the assistant works from real search-demand data. Typical sessions are short and interactive: a handful of queries to explore a topic, well inside Basic Access limits.

Architecture

The server is deliberately thin. It runs as a local process speaking MCP over stdio to a single client on the same machine, and is a pass-through to the Google Ads API: it fetches, validates and normalizes responses into stable typed shapes, and does nothing else. All interpretation — clustering, intent labelling, prioritisation — happens in the calling agent, never in the server.

- **Transport** — MCP over stdio; no network listener is opened and the server is not reachable from outside the machine.
- **API** — the official `google-ads` Python client, pinned to Google Ads API v25.
- **Scope** — one OAuth scope, `adwords`, against one configured customer ID.
- **Caching** — a small in-memory LRU cache (128 entries by default) de-duplicates repeated queries within a session, reducing redundant API calls. Nothing is written to disk.
- **Rate limiting** — requests are paced and retried with backoff so that bursts from an agent cannot overrun API quota.

Interface

The server exposes three read-only tools. None of them creates, modifies, or deletes anything in the Google Ads account; the server performs no campaign management and issues no mutate operations.

Tool	Purpose	Google Ads service
generate_keyword_ideas	Discover keywords from up to 20 seed terms, a single page URL, or a domain. Returns results ranked by search volume, with paging.	KeywordPlanIdeaService
get_keyword_historical_metrics	Enrich a known keyword list with average monthly searches, monthly volume history, paid competition, and top-of-page bid ranges.	KeywordPlanIdeaService

resolve_geo_targets	Resolve a plain-language location into exact Google Ads geo target resource names, so queries are scoped to the intended market.	GeoTargetConstantService

Data handling and presentation

Google Ads data is returned to the operator's own assistant for their own planning, and is not stored, published, or resold. Two presentation commitments are built into the design:

- **Paid competition is never relabelled.** The competition figures describe advertiser activity, and the server presents them as such. It does not rename them ‘difficulty’ or otherwise imply they measure organic ranking.
- **No composite scores.** The server computes no blended ‘keyword score’ that would obscure the provenance of Google Ads metrics. Figures are passed through as reported, with monetary values carried as exact amounts alongside their currency code.

Credentials are supplied through environment variables or a local configuration file, are never logged, and are excluded from version control. Configuration objects redact secrets in their representations so that credentials cannot leak into logs or error output.

Operating requirements

Item	Detail
Deployment	Local process on the operator's own machine; single user; not hosted or shared
Runtime	Python 3.10-3.14
Google Ads accounts	One customer ID, optionally accessed via a manager account
Authentication	OAuth 2.0 user credentials, or a service account
Expected volume	Interactive use only — tens of operations per session, far below Basic Access limits
Operations used	Read-only reporting and planning queries; no mutate operations

Keyword Research MCP — open-source, MIT licensed. This document describes the design and intended use of the tool.